APO-PIROXICAM
Piroxicam Capsules 20mg
Anti-inflammatory Agent with Analgesic Properties

Actions and Clinical Pharmacology: Apo-Piroxicam (Piroxicam) is a non-steroidal anti-inflammatory agent with analgesic and antipyretic properties. Its mechanism of action is not completely known. Piroxicam inhibits the activity of prostaglandin synthetase. The resulting decrease in prostaglandin biosynthesis may partially explain its anti-inflammatory action. Piroxicam does not act by pituitary adrenal stimulation. In rheumatoid arthritis the efficacy of piroxicam 20mg daily has been found to be similar to 4.2 g daily of ASA. Piroxicam is well absorbed following oral administration. The extent and rate of absorption are not influenced by administration with food or antacids. When the drug is administered daily, plasma concentrations increase for 5 to 7 days during which a steady state is reached. Concentrations attained are not exceeded following further constant daily drug intake.
A single dose (20mg) bioavailability study was performed by Richardson et al in order to evaluate age and sex related differences in the pharmacokinetics of piroxicam. It was found that elder women (62 to 75 years) had 33% lower piroxicam clearance than young women (20 to 31 years), a difference that was reflected in the significantly different half-life of 61.7 and 44.9 hours respectively. No significant difference of the clearance rate was found between young men and elderly men. The predicted steady state plasma concentrations were found to be 5.7, 5.4 and 5.7 mcg/mL in young women, young men and elderly men compared with 9.3 mcg/mL in elderly women. These results are at variance with the findings of Hobbs et al and Woolf et al, who concluded that age had no or negligible effect on piroxicam clearance and steady state plasma levels. Piroxicam is extensively metabolized and less than 5% of the daily dose is excreted unchanged in urine and feces. The main metabolic pathway is hydroxylation of the pyridyl ring, followed by conjugation with glucuronic acid and urinary elimination. Approximately 5% of the dose is metabolized to and excreted as saccharin. Over a 4 days period of observation, 20 healthy men taking piroxicam 20mg daily in single or divided doses, showed significantly less mean daily fecal blood loss than did 10 healthy male controls taking 3.9g of ASA daily.

Indications:

- For the symptomatic relief of pain and inflammation in patients with osteoarthritis, rheumatoid arthritis and ankylosing spondylitis.
- However it should not be the first choice of non-steroidal anti-inflammatory drug (NSAID) treatment in these conditions.

Contraindications: Apo-Piroxicam (piroxicam) is contraindicated in all patients with a recent or recurrent history of peptic ulceration or active inflammatory disease of the gastrointestinal tract. Piroxicam is contraindicated in those patients who have shown a hypersensitivity to the drug. Piroxicam should not be administered to patients in whom asthma, rhinitis or urticaria has been precipitated by ASA or other non steroidal anti-inflammatory agents because of the existence of cross sensitivity.

- Piroxicam should not be prescribed to patients who are more likely to develop side effects, such as those with history of gastrointestinal disorders associated with bleeding or those who have had skin reactions to other medicines.
- Piroxicam should not be prescribed in association with any other NSAID or an anticoagulant.

Warnings: Peptic ulceration, perforation, and GI bleeding-sometimes severe and in some instances fatal have been reported with patients receiving piroxicam. If piroxicam must be given to patients with a history of upper gastrointestinal tract disease, the patient should be under close supervision. (see contraindications). In controlled clinical trials, incidence of peptic ulceration with the maximum recommended piroxicam capsule dose of 20mg per day was 1%. The use of doses higher than the recommended dose is associated with an increase in the incidence of gastrointestinal irritation and ulcers. Therefore doses higher than 20mg should never be used. There is no evidence that the concurrent administration of H2 antagonists or antacids will either prevent the occurrence of gastrointestinal adverse reactions or allow continuation of piroxicam therapy in the presence of these adverse reactions.
There is some evidence that the incidence of serious adverse reactions is significantly increased in elderly, frail or debilitated patients. In these patients, treatment should be started with 10mg/day and increased to 20mg/day, if necessary. Special case should be taken to monitor these patients for adverse

reactions.
Risk of GI Ulceration, Bleeding and Perforation with NSAID: Serious GI toxicity such as bleeding, ulceration and perforation can occur at anytime with or without warning symptoms, in patients treated with NSAID therapy. Although minor GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms. In patients observed in clinical trials of several months to 2 years duration, symptomatic upper GI ulcers, gross bleeding or perforation occurred in approximately 1% of patients treated for 3 to 6 months and in about 2% to 4% of patients treated for 1 year. Higher percentages have been reported by other independent studies. Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, corticosteroid therapy) are at an increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events. Measures such as the use of physical therapy and mild analgesics like paracetamol (with inflammation is not a major factor) should be instituted prior to initiation of therapy with NSAID. NSAIDs should only be used after proper appraisal of potential risks to patients. It should be used with the lowest effective dose for only as long needed. This drug should not be co-administered with other NSAIDs. Prescribers should inform patients about the signs and/or symptoms of serious GI toxicity and what steps to take if they occur. In considering the use of relatively large doses (within the recommended dosage range) sufficient benefit should offset the potential increased risk of GI toxicity.

<u>Use in Pregnancy and Lactation:</u> The use of Apo-Piroxicam (piroxicam) during pregnancy or lactation is not recommended as its safety in these conditions has not been established. It is not known if piroxicam crosses the placental barrier or its excreted in breast milk. No teratogenic effects have been observed in animal reproductive studies with piroxicam. Rats and rabbits receiving piroxicam during pregnancy have shown an increased frequency of dystocia and delayed parturition. Rats have exhibited suppression of lactation.
<u>Use in Children:</u> Piroxicam is not recommended for use in children under 16 years of age as the dose and indications have not been established.

Precautions: As with other anti-inflammatory agents, long term administration to animals results in renal papillary necrosis and related pathology in rats, mice and dogs.
Acute renal failure and hyperkalemia as well as reversible elevations of BUN and serum creatinine have been reported with piroxicam. The effect is thought to result from inhibition of renal prostaglandin synthesis resulting in a change in medullary and deep cortical blood flow with an attendant effect on renal function. Patients with impaired renal function, congestive heart failure, cirrhosis with ascites or those on diuretics as well as elderly patients are more at risk.
Because of the extensive renal excretion of piroxicam and its biotransformation products (less than 5% of the daily dose excreted unchanged), lower doses of piroxicam should be anticipated in patients with impaired renal function and they should be carefully monitored. In addition to reversible changes in renal function, interstitial nephritis, glomerulitis, papillary necrosis and the nephrotic syndrome have been reported with piroxicam.
Peripheral edema has been observed in approximately 2% of the patients treated with piroxicam. Therefore, piroxicam should be used with caution in patients with heart failure, hypertension or other conditions predisposing to fluid retention. Congestive heart failure may be precipitated in the elderly and in patients with compromised cardiac function.
Although other nonsteroidal anti-inflammatory drugs do not have the same direct effects on platelets that aspirin does, all drugs inhibiting prostaglandin biosynthesis do interfere with platelet function to some degree; therefore patients who may be adversely affected by such an action should be carefully observed when piroxicam is administered.
Because of reports of adverse eye findings with nonsteroidal anti-inflammatory agents, it is recommended that patients who develop visual complaints during treatment with piroxicam have opthalmic evaluation.
With piroxicam borderline elevations of one or more liver tests may occur in up to 15% of patients. These abnormalities may progress and remain essentially unchanged, or may be transient with continued therapy. The SGPT (ALT) test is probably the most sensitive indicator of liver dysfunction. Meaningful (3 times the upper limit of normal) elevations of SGPT or SGOT (AST) occurred in controlled clinical trials in less than 1% of patients. A patient with symptoms and/or signs suggesting

liver dysfunction or in whom an abnormal liver test has occurred should be evaluated for evidence of the development of more severe hepatic reaction while on therapy with piroxicam. Severe hepatic reactions, including jaundice and cases of fatal hepatitis, have been reported with piroxicam. Although such reactions are rare, if abnormal liver tests persist or worsen, if clinical signs and symptoms consistent with liver disease develop, or if systemic manifestation occur (e.g. eosinophilia, rash, etc.), piroxicam should be discontinued. (See also Adverse Reactions).
Although at the recommended dose of 20mg/day of piroxicam increased fecal blood loss due to gastrointestinal irritation did not occur, in about 4% of the patients treated with piroxicam alone or concomitantly with ASA, reductions in hemoglobin and hematocrit values were observed. Therefore, these values should be checked periodically.
A combination of dermatological and/or allergic signs and symptoms suggestive of serum sickness have occasionally occurred in conjunction with the use of piroxicam. These include arthralgias, pruritus, fever, fatigue and rash including vesiculo-bullous reactions and dermatitis.
Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose and duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.
There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSIAD use.

- Treatment should always be initiated by a physician experienced in the treatment of rheumatic arthritis.
- Use the lowest dose (no more than 20 mg per day) and for the shortest duration possible. Treatment should be reviewed after 14 days.
- Always consider prescribing a gastroprotective agent.

Hypertension
NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired antihypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.

Heart Failure
Fluid retention and oedema have been observed in some patients taking NSAIDs; there caution is advised in patients with fluid retention or heart failure.

Gastrointestinal Events
All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patient about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

Severe Skin Reactions
NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens - Johnson syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reactions and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.

Drug Interactions: Piroxicam is highly protein bound, and, therefore, might be expected to displace other protein bound drugs. Although this has not occurred in the vitro studies with coumarin-type anticoagulants, interactions with coumarin type anticoagulants have been reported with piroxicam since marketing. Therefore, physicians should closely monitor patients for a change in dosage requirements when administering piroxicam to patients on coumarin-type anticoagulants and other highly protein bound drugs
Plasma level of piroxicam are depressed to approximately 80% of their normal values when piroxicam administered in conjunction with ASA (3900mg/day), but concomitant administration of antacids has no effect on piroxicam plasma levels.

Non-steroidal anti-inflammatory agents, including piroxicam, have been reported to increase steady state plasma lithium levels. It is recommended that plasma lithium levels be monitored when initiating, adjusting and discontinuing piroxicam.

Adverse Reactions: In patients receiving piroxicam, the most frequent side effects observed are gastrointestinal. It was reported that in a series of trials involving 1025 patients, the incidence of these reactions was 17.3%. 3.9% of patients required discontinuation of therapy. The most severe adverse reactions are peptic ulceration and gastrointestinal bleeding.
Adverse reactions reported in decreasing order of frequency are:
Gastrointestinal: Abdominal discomfort 5.7%, flatulence 5.2%, nausea 4.8%, abdominal pain 4.7%, epigastric distress 4.1%, constipation 3.8%, diarrhea 3.2%, peptic ulceration 1.8%, stomatitis 1-3%, anorexia 2.0%, vomiting 1.0%, indigestion 0.7%, gastrointestinal bleeding 0.1%. Liver function abnormalities, jaundice, hepatitis, hematemisis, melena, dry mouth were seen in less than 1% of patients. Central Nervous System: Dizziness 4.1%, headache 4.1%, drowsiness/sedation 2.1%. Amnesia, anxiety, depression, hallucinations, insomnia, nervousness, paresthesia, personality change, tremors and vertigo were seen in less than 1% of patients. Dermatologic: Rash 2.4%, pruritus 1.1%. Alopecia sweating, erythema, bruising, desquamation, exfoliative dermatitis, erythema multiforme, toxic epidermal necrolysis. Stevens-Johnson syndrome, vesiculo-bullous reaction and photoallergic reactions were seen in less than 1% of patients. Genito-Urinary: Edema 2.7%, BUN elevation 1-3%, creatinine elevation 1-3%, dysuria, urinary frequency, hematuria, oliguria, proteinurea, interstitial nephritis, renal failure, hyperkalemia, glomerulitis, papillary necrosis, nephrotic syndrome and menorrhagia were seen in less than 1% of patients. Eyes/Ears/Nose/Throat: Blurred vision, eye irritation/swelling, deafness, tinnitus, epistaxis, and glossitis were seen in less than 1% of patients. Hematological: Decreases in hemoglobin and hematocrit 3-9%, anemia 1-3%, leukopenia 1-3%, eosinophilia 1-3%. Thrombocytopenia, petechial rash, ecchymosis, bone marrow depression including aplastic anemia, and epistaxis were seen in less than 1% of patients. Cardiavascular/Respiratory: Hypertension, worsening of congestive heart failure, exacerbation of angina, palpitations, tachycardia and breathlessness were seen in less than 1% patients. Metabolic: Hypoglycemia, hyperglycemia, weight increase, and weight decrease were seen in less than 1% of patients. Hypersensitivity: Bronchospasm, anaphylaxis, angioedema, vasculitis, and serum sickness were seen in less than 1% of patients. Miscellaneous: Pain (colic), fever, flu-like syndrome, thirst, chills, flushing and increased appetite were seen in less than 1% of patients.

Symptoms and Treatment of Overdose: A few cases of overdoses (up in 1800mg) have been reported. Recovery was complete. Symptoms and signs included nausea, abdominal pain, multiple superficial gastric and duodenal ulcers, GIT bleeding.
No specific antidote is known to piroxicam. The usual supportive and symptomatic treatment is recommended.

Dosage and Administration: After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortness possible duration should be used.
In rheumatoid arthritis and ankylosing spondylitis it is recommended that therapy with Apo-Piroxicam (piroxicam) be initiated as a single daily dose of 20mg. If desired, this dose may be given as 10mg bid. Most patients will be maintained on 20mg daily. A relatively small number of patients may be maintained on 10mg daily.
In osteoarthritis the recommended starting dosage of Apo-Piroxicam (piroxicam) is 20mg daily. If desired this dose may be given as 10mg twice daily. The usual maintenance dose is 10-20 mg daily. Because of increased risk of toxicity for elderly, debilitated and frail patients or patients with impaired renal function, a starting dose of 10mg/day is recommended. The dose may be increased to 20mg/day, if necessary (see Warnings and Precautions).
Apo-Piroxicam (piroxicam) should not be given in doses greater than 20mg daily owing to an increased risk of gastrointestinal side effects.

Availability:
Apo-Piroxicam 20mg : Each hard gelatin capsules with maroon body and cap, filled with off-white powder. Printed "APO 20" on capsule shell contains piroxicam 20mg. Available in PVC Blister of 500 capsules.

Storage Conditions: Store below 25°C

Product Registration Holder: Pharmaforte (M) Sdn Bhd; 2, Jalan PJU 3/49, Sunway Damansara, 47810, Petaling Jaya, Selangor.

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Date of Revision of Package Insert: 26th Sep 19

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PT 0029C